Dynamic Compensators

Control Engineering (Reguleringsteknik)

Christoffer Sloth

chsl@mmmi.sdu.dk

SDU Robotics
The Maersk Mc-Kinney Moller Institute
University of Southern Denmark

Agenda

Introduction
- Curriculum

Anti-Windup
- Back-Calculation
- Conditional Integration

Dynamic Compensation
- Lead Compensation
- Lag Compensation

Introduction
Curriculum for Reguleringsteknik (REG)

Matematiske og grafiske metoder til syntese af lineære tidsinvariante systemer:[1]

- diskret og kontinuert tilstandsbeskrivelse
- analyse i tid og frekvens
- stabilitet, reguleringshastighed, følsomhed og fejl
- digitale PI, PID, LEAD og LAG regulatorer (serieregulatorer)
- tilstandsregulering, pole-placement og tilstands-estimering (observer)
- optimal regulering (least squares) og optimal tilstands-estimation (Kalman-filter)

Færdigheder:
Efter gennemførelse af kurset kan den succesfulde studerende:

- kunne analysere, dimensionere og implementere såvel kontinuert som tidsdiskret regulering af lineære tidsinvariante og stokastiske systemer

Kompetencer:
Efter gennemførelse af kurset kan den succesfulde studerende:

- anvende og implementere klassiske og moderne reguleringsteknikker for at kunne styre og regulere en robot hurtig og præcist

[1] Based on https://fagbesk.sam.sdu.dk/?fag_id=39673

Introduction
Course Overview

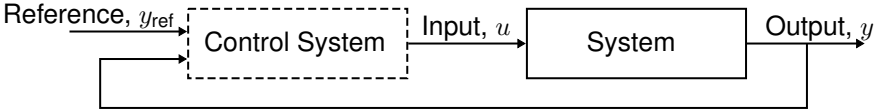
The twelve lectures of the course are

- **Lecture 1**: Introduction to Linear Time-Invariant Systems
- **Lecture 2**: Stability and Performance Analysis
- **Lecture 3**: Introduction to Control
- **Lecture 4**: Design of PID Controllers
- **Lecture 5**: Root Locus
- **Lecture 6**: The Nyquist Plot
- **Lecture 7**: Dynamic Compensators and Stability Margins
- **Lecture 8**: Implementation
- **Lecture 9**: State Feedback
- **Lecture 10**: Observer Design
- **Lecture 11**: Optimal Control (Linear Quadratic Control)
- **Lecture 12**: The Kalman Filter

Introduction

Motivating Example

Task: Design a cruise control for a car.



- **Control Input**: Throttle position u
- **Measured Output**: Velocity of the car y
- **Reference Input**: Desired velocity of the car y_{ref}

Introduction

Motivating Example

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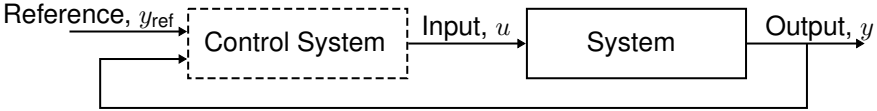

Today, an answer to the following question is provided:

1. How can one ensure that a control is stable despite uncertainties in the system?

Introduction
Motivating Example

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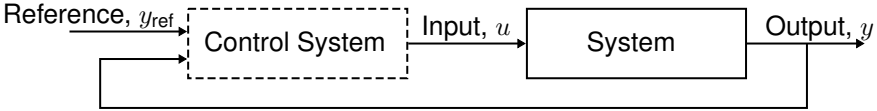


Today, an answer to the following question is provided:

1. How can one ensure that a control is stable despite uncertainties in the system?
   - Uncertainties may affect the *gain* of the system (e.g. the inertia).
   - Uncertainties may affect the *phase* of the system (e.g. communication delays between controller and sensor).

Introduction
Steady State Tracking

	Input		
Type	Step (Position)	Ramp (Velocity)	Parabola (Acceleration)
Type 0	$\frac{1}{1+K_p}$	∞	∞
Type 1	0	$\frac{1}{K_v}$	∞
Type 2	0	0	$\frac{1}{K_a}$

Anti-Windup

Introduction
Curriculum

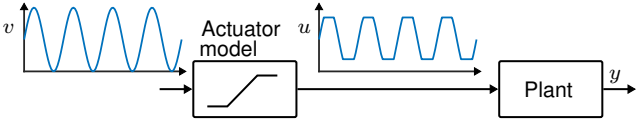
Anti-Windup
Back-Calculation
Conditional Integration

Dynamic Compensation
Lead Compensation
Lag Compensation

Anti-Windup

Introduction

Real electromechanical systems have saturations on their physical variables. In particular, the output of any actuator is limited from above and below.



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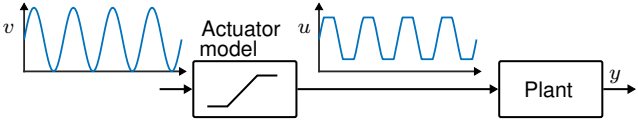


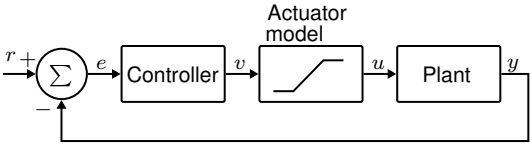
The saturation has the following input-output relation

$$u = \text{sat}(v) = \begin{cases} u_{\text{max}} & \text{if } v > u_{\text{max}} \\ u_{\text{min}} & \text{if } v < u_{\text{min}} \\ v & \text{otherwise} \end{cases}$$

Anti-Windup

Motivation

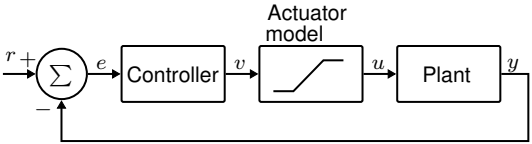
The saturation of actuators must be taken into account, when integral action is present in the controller.

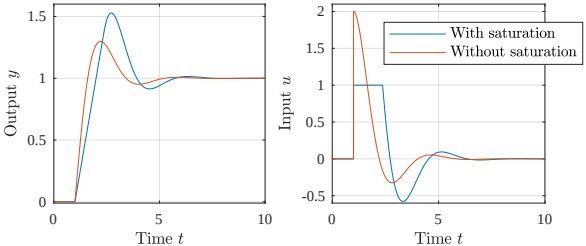


Anti-Windup

Motivation

The saturation of actuators must be taken into account, when integral action is present in the controller. Otherwise, the performance of the system may degrade significantly.

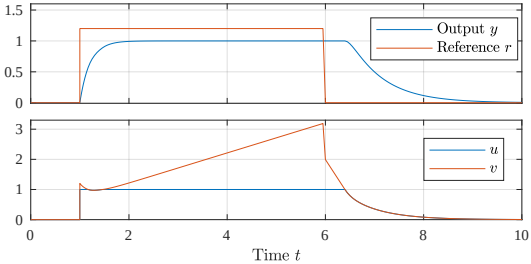




Anti-Windup

Setpoint Limits

If a setpoint is given to the system that cannot be reached, then the integrator winds up, and a delay is experienced when changing the setpoint value.



It is important to only give a system reachable references.

Anti-Windup
Overview

Two anti-windup mechanisms are considered

- Back-Calculation
- Conditional Integration

Anti-Windup
Overview

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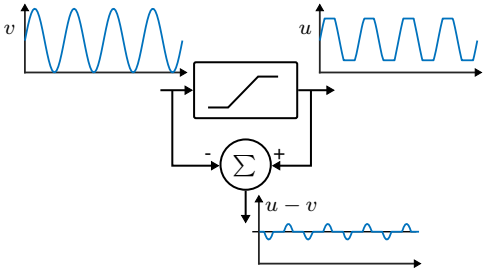
- Back-Calculation
- Conditional Integration

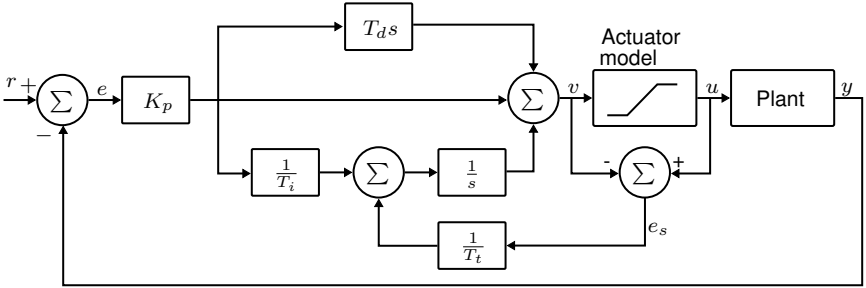
Both methods utilize the following signal



Anti-Windup

Back-Calculation

The anti-windup scheme called ***back-calculation*** recomputes the integral term when the system input is saturated, as shown below.



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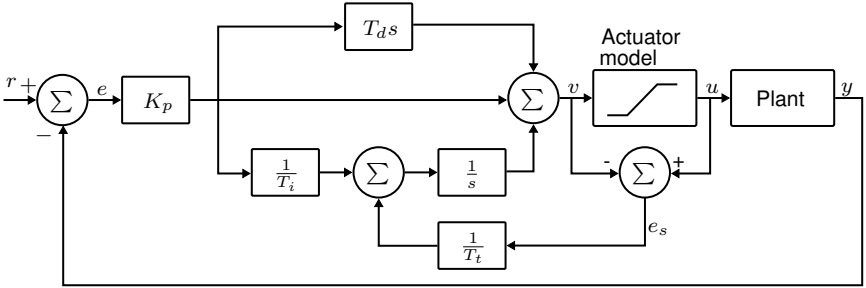

The value of the integrator output is not changed instantaneously, but it is changed based on the ***tracking time constant*** T_t.

Anti-Windup

Back-Calculation

The input to the integrator is given by

$$\frac{1}{T_t} e_s + \frac{K_p}{T_i} e$$

where e_s is zero when the system is not saturated.

In steady state, the output of the integrator is constant; hence, its input must be zero, i.e.

$$e_s = -\frac{K_p T_t}{T_i} e$$

in steady state.

Anti-Windup

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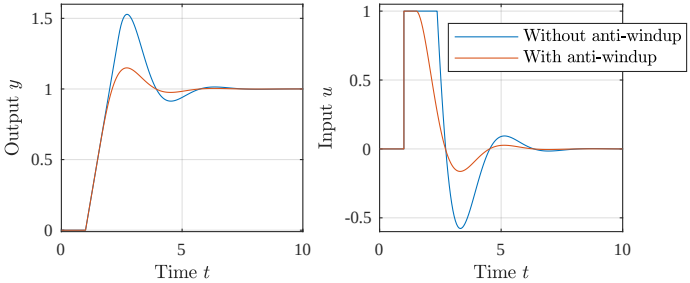
in steady state.Since $e_s = u - v$

$$v = u_{\text{lim}} + \frac{K_p T_t}{T_i} e$$

Anti-Windup

Back-Calculation

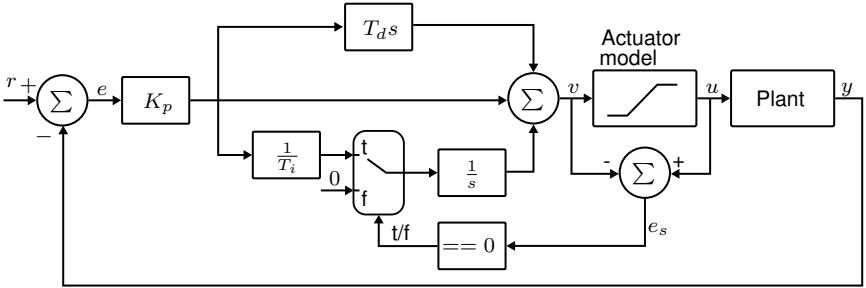
The use of anti-windup (here back-calculation) significantly improves the performance of the system.



Anti-Windup

Conditional Integration

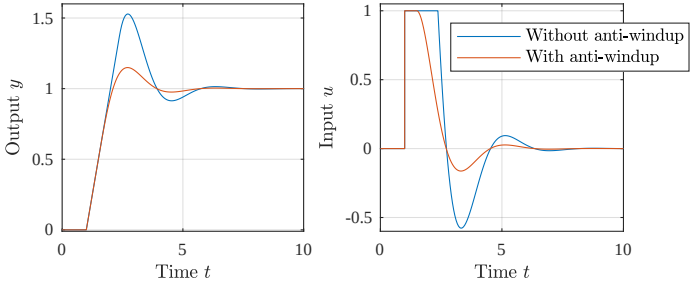
The anti-windup scheme called ***conditional integration*** (also known as clamping) is a bit simpler, and just stops integrating, when the system is in saturation.



Anti-Windup

Conditional Integration

Although conditional integration is simpler than back-calculation, it also improves the performance of the system.



Dynamic Compensation

Introduction
Curriculum

Anti-Windup
Back-Calculation
Conditional Integration

Dynamic Compensation
Lead Compensation
Lag Compensation

Dynamic Compensation
Overview

Two types of compensators are considered

- **Lead Compensation**: Approximates the PD control, i.e., it lowers the rise time and decreases the overshoot.
- **Lag Compensation**: Approximates the PI control, i.e., it improves the steady state tracking.

Dynamic Compensation
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Both compensators are given by the transfer function

$$D(s) = K\frac{s+z}{s+p}.$$

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Both compensators are given by the transfer function

$$D(s) = K\frac{s+z}{s+p}.$$

- If $z < p$, then $D(s)$ is called a lead compensation.
- If $z > p$, then $D(s)$ is called a lag compensation

Dynamic Compensation

Lead Compensation: Definition

A lead compensator is given by

$$D(s) = K\frac{s+z}{s+p}.$$

where $z < p \in \mathbb{R}$.

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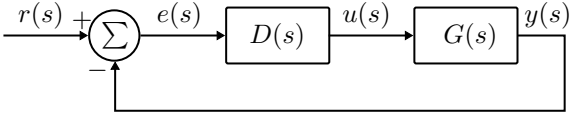
The transfer function $D(s)$ has a zero followed by a pole, and acts as a filtered PD controller, when used as a feedback controller.



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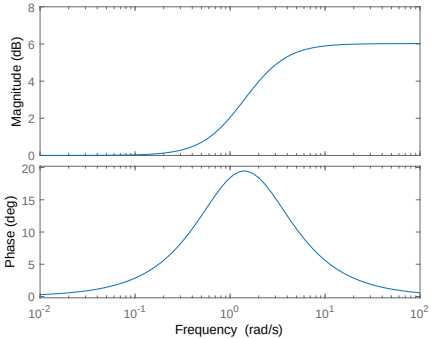

Dynamic Compensation

Lead Compensation: Parameters

A lead compensation is given by

$$D(s) = \frac{Ts + 1}{\alpha Ts + 1} \qquad \text{where } \alpha < 1$$

and $1/\alpha$ is called the ***lead ratio***.

Dynamic Compensation

Lead Compensation: Parameters

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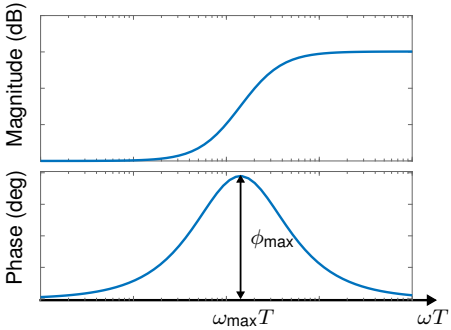

We have the following

$$\omega_{\text{max}} = \frac{1}{T\sqrt{\alpha}} = \sqrt{|z||p|}$$

and

$$\sin\phi_{\text{max}} = \frac{1-\alpha}{1+\alpha}$$

Dynamic Compensation

Lead Compensation: Design Procedure (Bode Plot)

1. Determine open-loop gain K to satisfy error or bandwidth requirements:
   1.1 To meet error requirement, pick K to satisfy error constants (K_p, K_V, K_a) so that e_{ss} error specification is met.
   1.2 Alternatively, to meet bandwidth requirement, pick K so that the open-loop crossover frequency is a factor of two below the desired closed-loop bandwidth, i.e., $\omega_{BW} = 2\omega_c$.
2. Evaluate the phase margin of the uncompensated system using the value K obtained from Step 1.
3. Allow for extra margin (about $10°$), and determine the needed phase lead ϕ_{max} (one lead compensation should contribute a maximum of $60°$ to the phase).
4. Determine α from

$$\sin \phi_{\text{max}} = \frac{1-\alpha}{1+\alpha}$$

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4. Determine α from
$$\sin\phi_{\text{max}} = \frac{1-\alpha}{1+\alpha}$$
5. Pick ω_{max} to be the crossover frequency; thus, the zero is at $1/T = \omega_{\text{max}}\sqrt{\alpha}$ and the pole is at $1/\alpha T = \omega_{\text{max}}/\sqrt{\alpha}$.
6. Draw the compensated frequency response and check the phase margin.
7. Iterate on the design. Adjust compensator parameters (poles, zeros, and gain) until all specifications are met.

Dynamic Compensation

Lead Compensation: Example (1)

Consider the system

$$KG(s) = \frac{K}{(s/0.5+1)(s+1)(s/2+1)}$$

Dynamic Compensation

Lead Compensation: Example (1)

Consider the system

$$KG(s) = \frac{K}{(s/0.5+1)(s+1)(s/2+1)}$$

Design a lead compensator such that the steady state error to a step input is

$$\frac{1}{1+K_p}$$

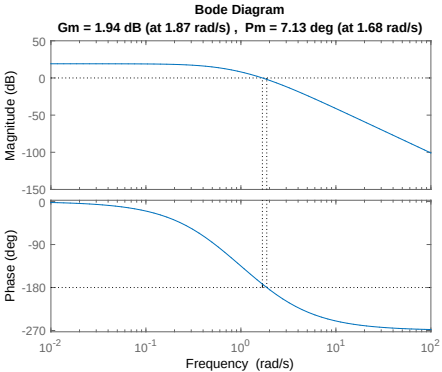
where $K_p = 9$ and the phase margin is at least $25°$.

Step 1: Meet steady state error requirement.

We use the Final Value Theorem to find K

$$K_p = \lim_{s \to 0} KG(s) = K = 9, \text{ since } G(0) = 1.$$

Step 2: The Bode plot of $KG(s)$ with $K = 9$ is shown below.



The phase margin is 7° at $\omega = 1.68$ rad/s.

Step 3: To allow a phase margin of 25° (requirement) plus 10° (extra margin), the needed phase lead is

$$\phi_{\text{max}} = 25^\circ + 10^\circ - 7^\circ = 28^\circ$$

Dynamic Compensation

Lead Compensation: Example (4)

Step 3: To allow a phase margin of 25° (requirement) plus 10° (extra margin), the needed phase lead is

$$\phi_{\text{max}} = 25^\circ + 10^\circ - 7^\circ = 28^\circ$$

Step 4: The value of α is computed as

$$\alpha = \frac{1 - \sin \phi_{\text{max}}}{1 + \sin \phi_{\text{max}}} = 0.3610$$

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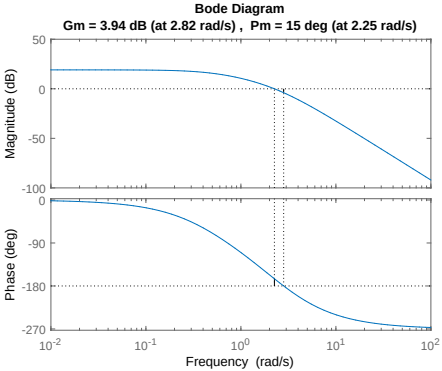
Step 5: Pick ω_{max} to be at the crossover frequency, i.e., $\omega_{\text{max}} = 1.68$ rad/s. Thereby,

$$T = \frac{1}{\omega_{\text{max}}\sqrt{\alpha}} = 0.9906$$

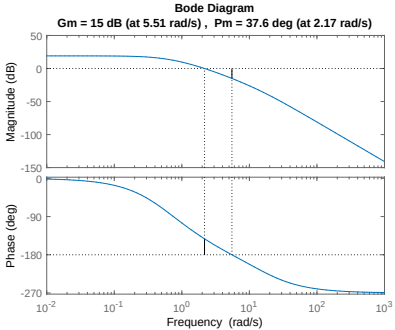
Step 6: Verify that the response satisfies the requirements. Since the crossover frequency has changed to 2.25 rad/s, the design does not work.



Step 7: The compensator is redesigned, as

$$D(s) = \frac{s/1.5 + 1}{s/15 + 1}$$

This gives the following frequency response



Dynamic Compensation

Lag Compensation: Definition

A lag compensator is given by

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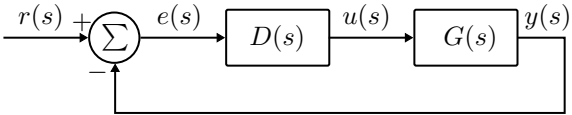
$$D(s) = K\frac{s+z}{s+p} \qquad or \qquad D(s) = K_0\alpha\frac{Ts+1}{\alpha Ts+1}$$

where $z > p \in \mathbb{R}$ and $\alpha > 1$.

The transfer function $D(s)$ has a pole followed by a zero, and approximates a PI controller, when used as a feedback controller.



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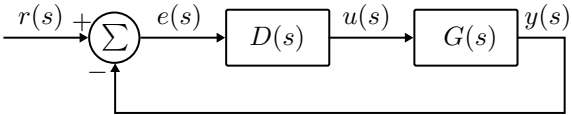

Idea: Improve the steady-state performance without affecting the other dynamics.

Dynamic Compensation

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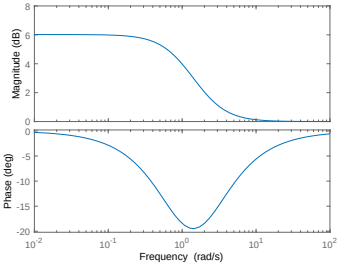

Dynamic Compensation

Lag Compensation: Design Procedure (Bode Plot)

1. Determine open-loop gain K that will meet the phase margin requirement without compensation.
2. Draw the Bode plot of the uncompensated system with crossover frequency from Step 1, and evaluate the low-frequency gain.
3. Determine α to meet the low-frequency gain error requirement.
4. Choose the corner frequency $\omega = 1/T$ (the zero of the lag compensator) to be one decade below the crossover frequency ω_c.
5. The other corner frequency (the pole location of the lag compensator) is $\omega = 1/\alpha T$
6. Iterate on the design. Adjust compensator parameters (poles, zeros, and gain) until all specifications are met.

Dynamic Compensation

Lag Compensation: Example (1)

Design a lag compensator so that the phase margin is at least 40° and $K_p = 9$ for the system

$$KG(s) = \frac{K}{(s/0.5+1)(s+1)(s/2+1)}$$

Dynamic Compensation

Lag Compensation: Example (1)

Design a lag compensator so that the phase margin is at least $40°$ and $K_p = 9$ for the system

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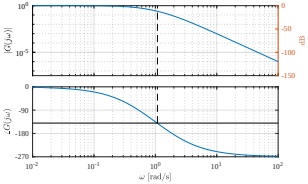
Step 1: Determine K such that $PM > 40°$.



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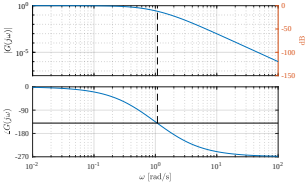
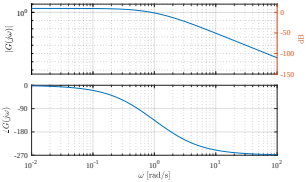
$$KG(s) = \frac{K}{(s/0.5+1)(s+1)(s/2+1)}$$

Step 1: Determine K such that $PM > 40°$.



We choose $K = 3$.

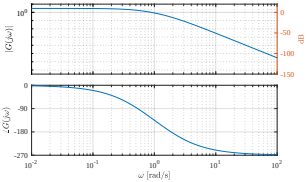
Step 2: Draw Bode plot, and evaluate low-frequency gain of $KG(s)$.



Dynamic Compensation

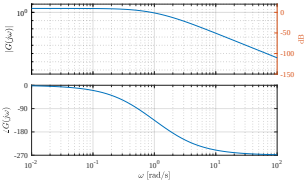
Lag Compensation: Example (2)

Step 2: Draw Bode plot, and evaluate low-frequency gain of $KG(s)$.



The phase margin is $53.5°$ and the low-frequency gain is 10 dB (or 3.16 times).

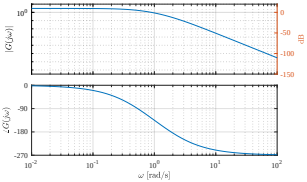
Step 2: Draw Bode plot, and evaluate low-frequency gain of $KG(s)$.



The phase margin is $53.5°$ and the low-frequency gain is 10 dB (or 3.16 times).

Step 3: Determine α such that desired low-frequency gain is attained.

Step 2: Draw Bode plot, and evaluate low-frequency gain of $KG(s)$.



The phase margin is $53.5°$ and the low-frequency gain is 10 dB (or 3.16 times).

Step 3: Determine α such that desired low-frequency gain is attained.

To get $K_p = 9$, α must be 3.

Dynamic Compensation

Lag Compensation: Example (3)

Step 4: Choose the zero of the lag compensator (corner frequency $\omega = 1/T$) to be one octave slower than the crossover frequency, i.e., $\omega = 0.2$ rad/s (and $T = 5$).

Dynamic Compensation

Lag Compensation: Example (3)

Step 4: Choose the zero of the lag compensator (corner frequency $\omega = 1/T$) to be one octave slower than the crossover frequency, i.e., $\omega = 0.2$ rad/s (and $T = 5$).
Step 5: The corner frequency of the pole of the lag compensator should be chosen to be $\omega = 1/\alpha T$. Thereby, the lag compensator is given by

$$D(s) = 3\frac{5s+1}{15s+1}$$

Dynamic Compensation

Lag Compensation: Example (3)

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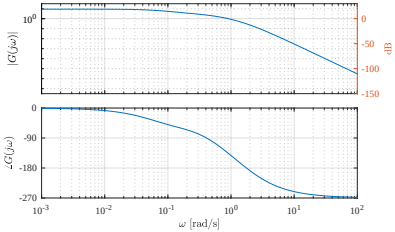

The system complies with the specification.